

ORIGINAL RESEARCH

Post-Operative Opioid Use After Total Hip & Knee Arthroplasty: The Role of Pre-Surgical Diabetes Management Strategy

Study 2 — Version 2 | April 2026 | Target: *BMC Musculoskeletal Disorders*

Method: Firth penalised logistic regression, bootstrap 95% CI (1,000 iter). Reference arm: Metformin. GLP-1 agonists merged into OtherDM arm.

1. Study Overview

This retrospective cohort study examines whether pre-surgical diabetes pharmacotherapy class is independently associated with post-operative opioid use following elective total hip arthroplasty (THA) or total knee arthroplasty (TKA). Opioid exposure after joint replacement is a major driver of morbidity, dependency risk, and healthcare cost. Understanding whether glycaemic management strategy influences opioid burden may identify a modifiable, pre-surgical lever for reducing post-operative opioid utilisation.

Using the same 976-patient EHR cohort as the companion readmission study (Study 1), patients were assigned to one of four arms: **Metformin** (reference), **Insulin**, **OtherDM**, and **DietControlled**. The primary outcome is any opioid dispensation within 90 days of surgery. A critical covariate unique to this study is **baseline opioid use** (pre-surgical opioid exposure), the strongest known predictor of post-operative opioid use. Eight sensitivity analyses test robustness across windows, outcome definitions, and patient subgroups including opioid-naïve and opioid-experienced patients.

Key Design Decisions

Decision	Rationale
Reference arm: Metformin	First-line T2DM therapy with largest pharmacological arm (n=125). Clinically meaningful comparator.
GLP-1 merged into OtherDM	GLP-1 users distributed across arms (n=218). Merging preserves statistical power.
Firth penalised logistic regression	Corrects separation bias in small OtherDM arm (n=47). More reliable ORs than standard logistic regression.
Baseline opioid use as covariate	57.9% of cohort had pre-surgical opioid exposure. Without adjustment, any arm effect is confounded by prior opioid habits — the strongest predictor of post-operative use.
Primary outcome: Any opioid 90d	Claims-based, objective endpoint capturing the full post-operative opioid exposure period including discharge prescriptions and refills.
8 sensitivity analyses	Post-operative opioid use is heavily influenced by prior exposure, surgery type, and outcome definition. Extensive sensitivity testing is required to isolate the DM arm signal.

2. Cohort Construction

Identical cohort to Study 1: 1,953 raw → 49 excluded (T1DM) → 128 excluded (insufficient baseline) → **976 final patients**.
Notably, baseline opioid use is markedly higher in the Insulin arm (76.6%) compared to Metformin (52.8%) and DietControlled (52.1%), reflecting the higher comorbidity burden in insulin-dependent patients.

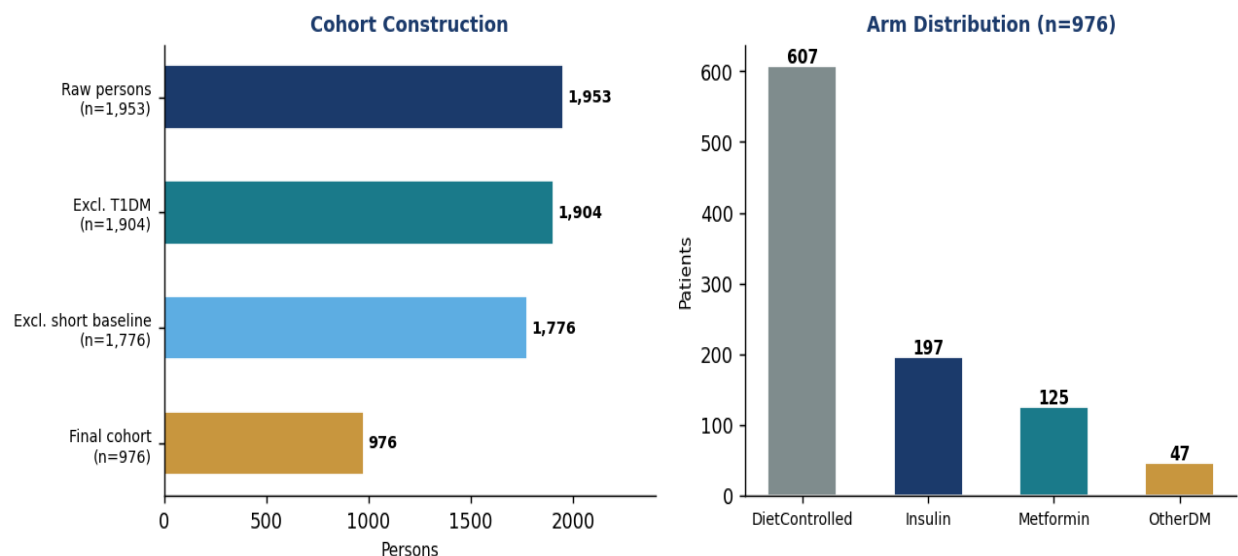


Figure 1. Left: Cohort exclusion funnel. Right: Final arm distribution.

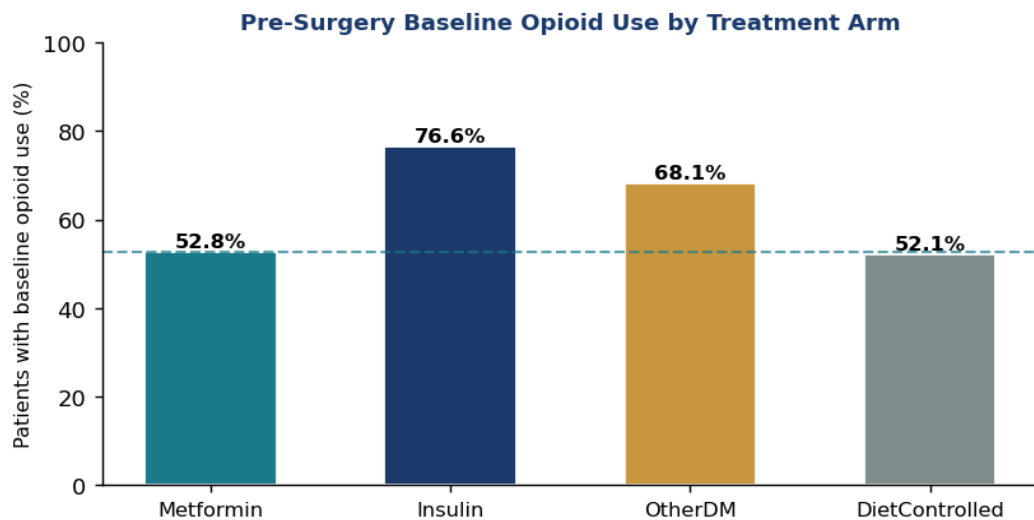


Figure 2. Pre-surgical baseline opioid use by arm. Dashed = Metformin (52.8%). Insulin arm markedly elevated (76.6%).

3. Study Variables — Complete Reference

All variables considered in this study are catalogued below with operational definitions. The key addition versus Study 1 is **baseline opioid use** as a model covariate.

Variable	Type	Role	Availability	Definition / Notes
Any Opioid 90d	Binary	Primary outcome	100%	Any opioid dispensation within 90 days of index THA/TKA. Identified from drug records with opioid drug class coding.
Any Opioid 30d	Binary	Sensitivity outcome	100%	Any opioid dispensation within 30 days. Captures the immediate post-discharge period.
Prolonged Opioid ≥ 2 Rx 90d	Binary	Sensitivity outcome	100%	Two or more distinct opioid dispensations within 90 days. Proxy for sustained/prolonged post-operative use — the highest risk threshold for opioid dependency.
DM Treatment Arm	Categorical (4 levels)	Main Predictor	100%	Assigned from drug dispensation claims in 365d prior to surgery. Hierarchy: Insulin > OtherDM > Metformin > DietControlled.
Baseline Opioid Use	Binary (yes/no)	KEY Covariate	100% (57.9% positive)	Any opioid dispensation in the 365-day baseline window. Present in 565/976 patients. The strongest known predictor of post-operative opioid use; must be included to isolate the DM arm effect. Rates by arm: Metformin 52.8%, Insulin 76.6%, OtherDM 68.1%, DietControlled 52.1%.
Age at surgery	Continuous (yrs)	Covariate	100%	Patient age on date of index procedure.
BMI	Continuous (kg/m ²)	Covariate	54.5% (n=532)	Most recent pre-surgery BMI from vital-sign measurement records.
HbA1c	Continuous (%)	Covariate	46.9% (n=458)	Most recent pre-surgery glycated haemoglobin. Available-case analysis for missing values.
Diabetes Duration	Continuous (yrs)	Covariate	100%	Time from first DM diagnosis code to surgery date.
Surgery Type	Binary (THA/TKA)	Covariate	100%	Index procedure. Significant surgery x Insulin interaction found (8G, p=0.041) — TKA-specific Insulin effect.
Hypertension	Binary	Covariate	100%	ICD code for hypertension in 365-day baseline.
Hyperlipidaemia	Binary	Covariate	100%	ICD code for hyperlipidaemia in 365-day baseline.
CAD	Binary	Covariate	100%	Coronary artery disease in baseline condition records.
Depression	Binary	Covariate	100%	Depression diagnosis in baseline. Included as known correlate of opioid use.
Baseline Inpatient Visits	Continuous (count)	Covariate	100%	Inpatient encounters in 365-day baseline. Proxy for overall comorbidity burden.
Outpatient Visits	Continuous (count)	Descriptive	100%	Outpatient encounter count in baseline. Reported in Table 1 only.
ED Visits	Continuous (count)	Descriptive	100%	ED visit count in baseline. Reported in Table 1 only.

Variable	Type	Role	Availability	Definition / Notes
GLP-1 Agonist Use	Binary	Note	100%	Identified within each arm (n=218 total). Merged into OtherDM in v2.

Blue = outcomes. Orange = main predictor. Gold (bold) = key covariate (baseline opioid use).

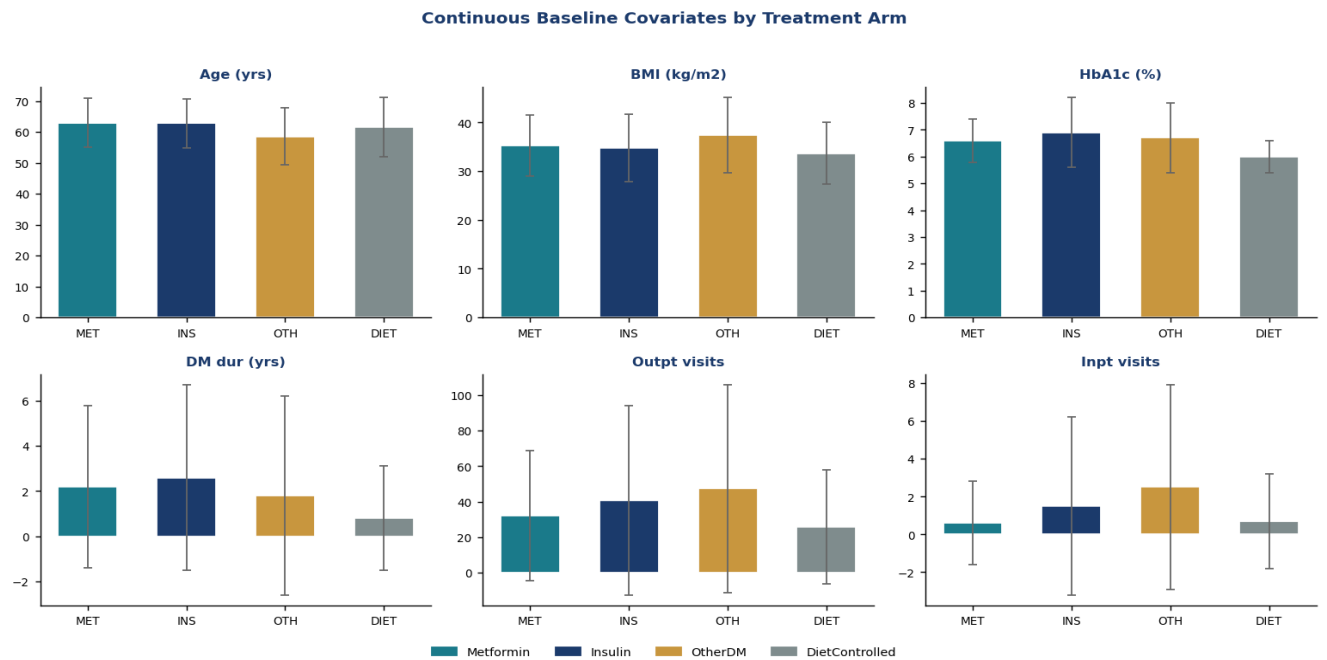


Figure 3. Continuous baseline covariates (mean ± SD) by arm.

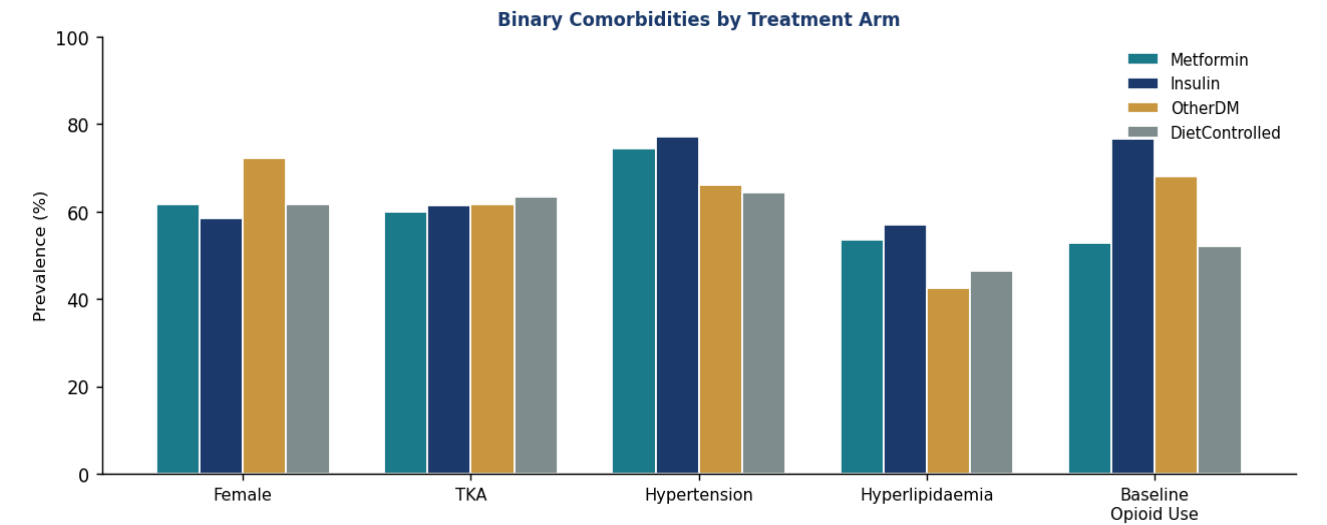


Figure 4. Binary comorbidities and baseline opioid use by arm.

4. Statistical Tests Performed

This study deploys 8 sensitivity analyses — more than Study 1 — because post-operative opioid use is strongly influenced by prior exposure, surgery type, and outcome definition.

Test / Model	Outcome(s)	Rationale
Firth Penalised Logistic (primary, bootstrap CI)	opioid_90d	Corrects separation bias; reliable ORs with bootstrap CIs. Controls for baseline opioid use, age, BMI, HbA1c, DM duration, surgery type, HTN, HLD, CAD, depression, inpatient visits.
GLM HC3 Robust SE (cross-check)	opioid_90d	Independent validation. Concordance between Firth and GLM strengthens inference.
8A: 180-day window	opioid_90d	Verifies arm assignments are not artefacts of the 365-day look-back definition.
8B: Surgery type subgroup (THA / TKA)	opioid_90d	THA and TKA differ in pain profile and opioid prescribing norms. Tests procedure-specific effects.
8C: Active pharma only (excl. DietControlled)	opioid_90d	Isolates drug-class contrasts among patients on confirmed pharmacological treatment.
8D: 30-day opioid	opioid_30d	Captures the immediate post-discharge period (acute pain management) vs the full 90-day window.
8E: Prolonged opioid (≥ 2 Rx / 90d)	prolonged_90d	Clinically meaningful threshold for dependency risk. Tests whether OtherDM elevation persists for sustained rather than one-time use.
8F: Opioid-naïve only (n=411)	opioid_90d	Restricts to patients with no prior opioid exposure. De novo initiation = highest clinical risk. Removes pre-existing opioid priming as a confound.
8G: Surgery type x arm interaction test	opioid_90d	Formal test of whether surgery type modifies the DM arm effect. Significant interaction ($p < 0.05$) means subgroup results must be reported separately rather than pooled.
8H: Opioid-experienced only (n=565)	opioid_90d	Tests whether the arm effect persists in patients already habituated to opioids, where chronic pain contributions are harder to disentangle from surgical effects.

5. Table 1 — Baseline Characteristics

Key observations: Insulin arm has the highest baseline opioid rate (76.6%) and inpatient visit count (1.5). OtherDM has the highest BMI (37.4) and most outpatient visits (47.3). DietControlled has the lowest baseline opioid rate (52.1%) and the lowest primary outcome rate (59.5%), contrasting sharply with OtherDM (91.5%).

Variable	Metformin (n=125)	Insulin (n=197)	OtherDM (n=47)	DietCtrl (n=607)
Age (yrs)	63.0 $\pm$ 8.0	62.8 $\pm$ 7.9	58.5 $\pm$ 9.2	61.6 $\pm$ 9.5
BMI (kg/m2)	35.3 $\pm$ 6.3	34.8 $\pm$ 6.9	37.4 $\pm$ 7.7	33.7 $\pm$ 6.4
HbA1c (%)	6.6 $\pm$ 0.8	6.9 $\pm$ 1.3	6.7 $\pm$ 1.3	6.0 $\pm$ 0.6
DM Duration (yrs)	2.2 $\pm$ 3.6	2.6 $\pm$ 4.1	1.8 $\pm$ 4.4	0.8 $\pm$ 2.3
Outpatient visits	32.1 $\pm$ 36.8	40.7 $\pm$ 53.2	47.3 $\pm$ 58.4	25.8 $\pm$ 32.1
Inpatient visits	0.6 $\pm$ 2.2	1.5 $\pm$ 4.7	2.5 $\pm$ 5.4	0.7 $\pm$ 2.5
ED visits	0.6 $\pm$ 2.5	0.8 $\pm$ 2.0	0.2 $\pm$ 0.5	0.5 $\pm$ 1.8
Female	77 (61.6%)	115 (58.4%)	34 (72.3%)	375 (61.8%)
TKA	75 (60.0%)	121 (61.4%)	29 (61.7%)	385 (63.4%)

Variable	Metformin (n=125)	Insulin (n=197)	OtherDM (n=47)	DietCtrl (n=607)
Hypertension	93 (74.4%)	152 (77.2%)	31 (66.0%)	390 (64.3%)
Hyperlipidaemia	67 (53.6%)	112 (56.9%)	20 (42.6%)	282 (46.5%)
Baseline opioid use	66 (52.8%)	151 (76.6%)	32 (68.1%)	316 (52.1%)
— OUTCOMES —				
Any Opioid 90d (primary)	94 (75.2%)	158 (80.2%)	43 (91.5%)	361 (59.5%)
Any Opioid 30d	83 (66.4%)	150 (76.1%)	39 (83.0%)	336 (55.4%)
Prolonged ≥ 2 Rx 90d	72 (57.6%)	137 (69.5%)	33 (70.2%)	285 (47.0%)

Baseline opioid use (orange) = model covariate. Outcome rows (red) = unadjusted rates.

6. Unadjusted Outcome Rates

All three opioid outcomes follow the same rank order: OtherDM > Insulin > Metformin > DietControlled. Adjusted analysis follows in Section 7.

Outcome	Arm	Events	N	Rate
Any Opioid 90d (primary)	DietControlled Insulin	361 158 94	607 197	59.5% 80.2%
	Metformin OtherDM	43	125 47	75.2% 91.5%
Any Opioid 30d (sensitivity)	DietControlled Insulin	336 150 83	607 197	55.4% 76.1%
	Metformin OtherDM	39	125 47	66.4% 83.0%
Prolonged Opioid ≥ 2 Rx 90d	DietControlled Insulin	285 137 72	607 197	47.0% 69.5%
	Metformin OtherDM	33	125 47	57.6% 70.2%

Unadjusted Post-Op Opioid Event Rates by Arm

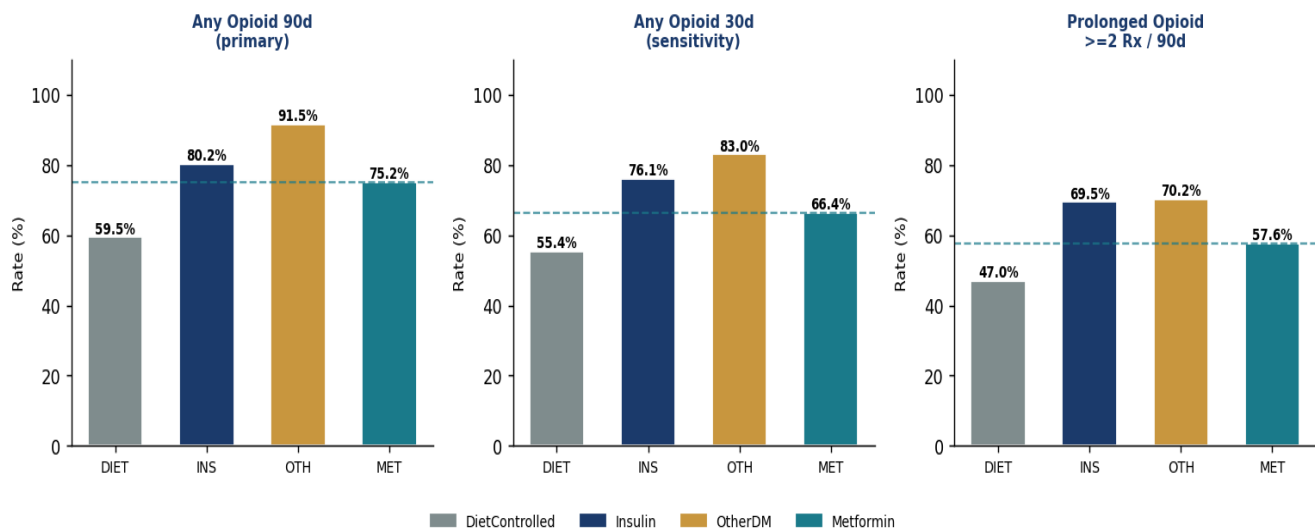


Figure 5. Unadjusted opioid rates. Dashed = Metformin reference. DietControlled is lowest across all three definitions.

7. Adjusted Analysis — Primary Outcome

Adjusted ORs from Firth penalised logistic regression and GLM HC3 cross-check. **Red** = elevated and significant. **Green** = protective and significant. Reference = Metformin.

Comparison	Firth aOR [95% CI]	p (Firth)	GLM aOR [95% CI]	p (GLM)	Direction	Sig?
Insulin vs MET	1.28 [0.64-2.47]	0.432	1.29 [0.68-2.47]	0.434	Higher	No
OtherDM vs MET	3.53 [1.31-14.13]	0.012	4.33 [1.18-15.92]	0.028	Much higher	YES
DietCtrl vs MET	0.30 [0.16-0.48]	0.001	0.29 [0.17-0.50]	0.000	PROTECTIVE	YES

Key findings: (1) **OtherDM:** 3.5x higher adjusted odds of any 90-day opioid (Firth aOR 3.53, $p=0.012$; GLM aOR 4.33, $p=0.028$). Concordant between methods. (2) **DietControlled:** Highly significant protective association — 70% lower odds (aOR 0.30, $p=0.001$). Opposite direction to readmission findings. May reflect lower pre-surgical pain burden, different prescribing patterns, or milder disease severity. (3) **Insulin:** Non-significant overall (aOR 1.28, $p=0.432$) — the high raw opioid rate in the Insulin arm is explained by its high baseline opioid use (76.6%), fully captured by the covariate.

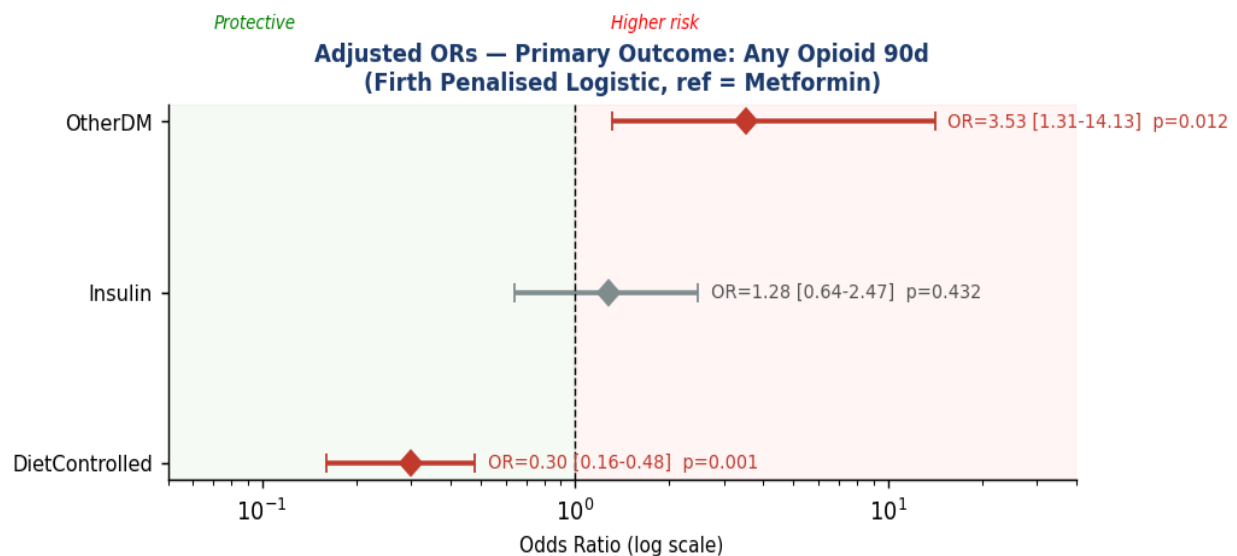


Figure 6. Forest plot (log scale). Green shading = protective region ($OR < 1$). Red = significantly elevated.

8. Sensitivity Analyses

8A — 180-Day Exposure Window

Consistent results with narrower look-back. OtherDM remains significant (aOR 4.25, $p=0.006$), DietControlled remains protective (aOR 0.36, $p=0.001$), Insulin non-significant ($p=0.278$).

Comparison	aOR [95% CI]	p	Sig?
Insulin vs MET	1.44 [0.76-2.85]	0.278	No
OtherDM vs MET	4.25 [1.58-18.83]	0.006	YES
DietCtrl vs MET	0.36 [0.19-0.60]	0.001	YES (protective)

8B — Subgroup by Surgery Type (THA vs TKA)

OtherDM significant and stronger in TKA (aOR 5.20, $p=0.008$). THA does not reach significance (aOR 2.64, $p=0.136$; limited power, $n=18$ OtherDM in THA). In TKA, Insulin also becomes significant (aOR 2.88, $p=0.012$) — not seen in pooled analysis. DietControlled protective in both THA ($p=0.002$) and TKA ($p=0.001$).

Surgery	Comparison	aOR [95% CI]	p	Sig?
THA	Insulin vs MET	0.71 [0.28-1.68]	0.448	No
THA	OtherDM vs MET	2.64 [0.73-12.62]	0.136	No
THA	DietCtrl vs MET	0.34 [0.14-0.69]	0.002	YES (protective)
TKA	Insulin vs MET	2.88 [1.30-7.54]	0.012	YES
TKA	OtherDM vs MET	5.20 [1.42-14.58]	0.008	YES
TKA	DietCtrl vs MET	0.29 [0.12-0.53]	0.001	YES (protective)

8C — Active Pharmacotherapy Only

Excluding DietControlled: OtherDM significant (aOR 3.53, $p=0.012$), Insulin non-significant ($p=0.432$). Identical to primary analysis conclusions.

Comparison	aOR [95% CI]	p	Sig?
Insulin vs MET	1.28 [0.64-2.47]	0.432	No
OtherDM vs MET	3.53 [1.31-14.13]	0.012	YES

8D — 30-Day Opioid Use

OtherDM remains significant (aOR 2.55, $p=0.022$); DietControlled remains protective (aOR 0.43, $p=0.001$); Insulin borderline (aOR 1.71, $p=0.094$). OtherDM attenuation at 30d (2.55 vs 3.53 at 90d) suggests risk accumulates over the full 90-day period.

Comparison	aOR [95% CI]	p	Sig?
Insulin vs MET	1.71 [0.93-3.21]	0.094	No (borderline)
OtherDM vs MET	2.55 [1.18-8.42]	0.022	YES
DietCtrl vs MET	0.43 [0.26-0.69]	0.001	YES (protective)

8E — Prolonged Opioid (>=2 Prescriptions / 90 Days)

OtherDM loses significance for prolonged use (aOR 1.89, p=0.120). Insulin approaches significance (aOR 1.75, p=0.056). DietControlled remains protective (aOR 0.46, p=0.004). Suggests OtherDM patients initiate but may not sustain opioid use post-operatively.

Comparison	aOR [95% CI]	p	Sig?
Insulin vs MET	1.75 [0.99-2.99]	0.056	No (borderline)
OtherDM vs MET	1.89 [0.84-5.00]	0.120	No
DietCtrl vs MET	0.46 [0.26-0.74]	0.004	YES (protective)

8F — Opioid-Naive Patients Only (n=411)

Restricting to patients with no pre-surgical opioid exposure isolates de novo initiation — highest clinical risk. OtherDM signal strengthens (aOR 4.34, $p=0.004$); DietControlled protection intensifies (aOR 0.22, $p=0.001$). Insulin non-significant ($p=0.818$). Confirms arm effects are not simply driven by pre-existing opioid habits.

Comparison	aOR [95% CI]	p	Sig?
Insulin vs MET	1.15 [0.41-2.83]	0.818	No
OtherDM vs MET	4.34 [1.65-10.68]	0.004	YES
DietCtrl vs MET	0.22 [0.10-0.44]	0.001	YES (protective)

8G — Surgery Type x DM Arm Interaction Test

Tests whether surgery type modifies the DM arm-opioid association. Significant interaction ($p<0.05$) means THA and TKA subgroup estimates must be reported separately.

Comparison	Interaction p	Sig. Interaction?	Interpretation
Insulin vs MET	0.041	YES ***	Effect differs by surgery type. TKA: aOR 2.88 ($p=0.012$); THA: aOR 0.71 ($p=0.448$). Report Insulin results separately by procedure type.
OtherDM vs MET	0.561	No	OtherDM effect consistent across procedures. Pooled primary estimate appropriate.
DietCtrl vs MET	0.660	No	DietControlled protective effect consistent across both surgery types.

8H — Opioid-Experienced Patients Only (n=565)

OtherDM persists but attenuated (aOR 3.33, $p=0.044$). DietControlled loses significance (aOR 0.56, $p=0.136$), suggesting its primary analysis protection is partly mediated by lower pre-operative opioid priming. Insulin non-significant ($p=0.206$).

Comparison	aOR [95% CI]	p	Sig?
Insulin vs MET	1.63 [0.74-3.65]	0.206	No
OtherDM vs MET	3.33 [1.06-14.12]	0.044	YES
DietCtrl vs MET	0.56 [0.22-1.13]	0.136	No (attenuated)

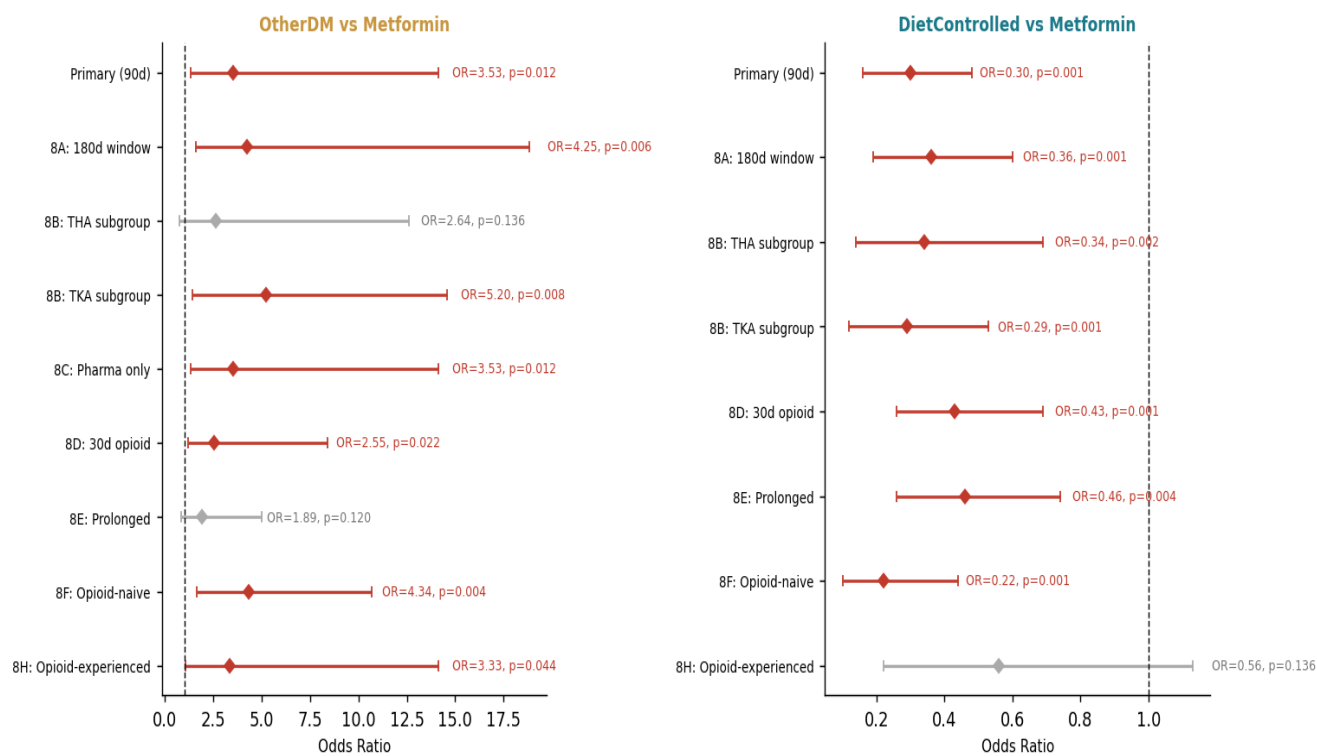


Figure 7. Sensitivity forest plots: OtherDM (left) and DietControlled (right). Red = significant. OtherDM consistently elevated; DietControlled consistently protective except in opioid-experienced subgroup (8H).

9. Results Summary

Analysis	Insulin aOR [CI] p	OtherDM aOR [CI] p	DietControlled aOR [CI] p
Primary: Opioid 90d	1.28 [0.64-2.47] p=0.432 ✗	3.53 [1.31-14.13] p=0.012 ✓	0.30 [0.16-0.48] p=0.001 ✓↓
GLM cross-check	1.29 [0.68-2.47] p=0.434 ✗	4.33 [1.18-15.92] p=0.028 ✓	0.29 [0.17-0.50] p=0.000 ✓↓
8A: 180d window	1.44 [0.76-2.85] p=0.278 ✗	4.25 [1.58-18.83] p=0.006 ✓	0.36 [0.19-0.60] p=0.001 ✓↓
8B: THA	0.71 [0.28-1.68] p=0.448 ✗	2.64 [0.73-12.62] p=0.136 ✗	0.34 [0.14-0.69] p=0.002 ✓↓
8B: TKA	2.88 [1.30-7.54] p=0.012 ✓	5.20 [1.42-14.58] p=0.008 ✓	0.29 [0.12-0.53] p=0.001 ✓↓
8D: 30d opioid	1.71 [0.93-3.21] p=0.094 ~	2.55 [1.18-8.42] p=0.022 ✓	0.43 [0.26-0.69] p=0.001 ✓↓
8E: Prolonged	1.75 [0.99-2.99] p=0.056 ~	1.89 [0.84-5.00] p=0.120 ✗	0.46 [0.26-0.74] p=0.004 ✓↓
8F: Opioid-naïve	1.15 [0.41-2.83] p=0.818 ✗	4.34 [1.65-10.68] p=0.004 ✓	0.22 [0.10-0.44] p=0.001 ✓↓
8H: Opioid-experienced	1.63 [0.74-3.65] p=0.206 ✗	3.33 [1.06-14.12] p=0.044 ✓	0.56 [0.22-1.13] p=0.136 ✗

✓ = $p < 0.05$ both methods ✓↓ = protective ($OR < 1$) and significant ~ = borderline ✗ = not significant

10. Outcome Definition — Clarification on Readmission Question

Question: In Study 1, are the 90-day readmissions (co-primary) admissions directly related to the surgery, versus the 90-day readmission or ER (secondary)?

In this EHR/claims-based study, **90-day readmission is defined as all-cause** — it captures *any* inpatient hospital admission within 90 days of the index THA/TKA, whether or not it was directly attributable to the surgical procedure. This means the readmission endpoint includes surgery-related events (wound infection, prosthetic dislocation, venous thromboembolism, post-operative fracture) *and* unrelated events (DM decompensation, cardiac events, urinary tract infection). In administrative and EHR-based research, restricting readmissions to surgery-related causes requires manual adjudication or procedure-specific diagnosis code validation that was not performed in this study.

The **90-day readmission or ED visit (secondary composite)** is broader: it adds all emergency department visits that did not result in inpatient admission. Many post-surgical complications or DM-related acute events are managed in the ED and the patient is discharged without formal admission — these would be invisible to the readmission-only endpoint. The composite therefore captures the *full spectrum of unplanned acute-care utilisation* following surgery.

Endpoint	Scope	What it includes	Clinical interpretation
90-Day Readmission (co-primary)	Inpatient admissions only ≤90 days post-surgery	All-cause inpatient admissions; NOT restricted to surgical causes	High-severity tier requiring hospital-level care. Includes surgical AND non-surgical causes.
30-Day Readmission (co-primary)	Inpatient admissions only ≤30 days post-surgery	Same as above; narrower window enriches for surgery-related events	Narrower window; higher likelihood of surgical causation but still all-cause.
90-Day Readmit or ED (secondary composite)	Inpatient admissions + ED visits ≤90 days	All inpatient admissions AND ED visits not resulting in admission	Full unplanned acute-care burden. Captures moderate-severity events managed in ED.

11. Study Limitations

The following limitations should be considered when interpreting findings and planning future research.

Limitation	Description	Impact on Interpretation
Retrospective observational design	Arm assignment is based on pre-existing drug dispensation, not randomisation. Unmeasured confounders (pain sensitisation, neuropathy severity, surgeon prescribing habits, anaesthesia type) cannot be excluded.	Cannot establish causality. Findings are hypothesis-generating. Randomised or propensity-matched designs are needed to confirm.
Opioid use as a proxy outcome	Any opioid dispensation is captured from drug records, not verified prescription fill or actual consumption. Some patients may have been dispensed opioids that were not used.	Outcome may slightly overestimate true opioid use. Under-ascertainment of informal opioid supply is also possible.
All-cause opioid endpoint	The outcome captures any opioid dispensation, regardless of whether it was prescribed for surgical pain or a pre-existing chronic pain condition.	In patients with chronic pain (particularly the OtherDM and Insulin arms), post-operative opioids may partly reflect continuation of pre-surgical therapy rather than new surgical pain treatment.
Small OtherDM arm (n=47)	Limited statistical power, particularly in subgroup analyses. Wide confidence intervals, especially in THA subgroup (n=18 OtherDM).	OtherDM findings should be replicated in larger cohorts. The THA non-significance likely reflects power, not absence of effect.
Incomplete HbA1c (47%) and BMI (55%)	Both covariates were available for less than 55% of patients. Available-case analysis used.	Potential bias if missingness is related to arm assignment. Residual confounding by glycaemic control is possible.
GLP-1 agonists merged into OtherDM	Prevents isolation of GLP-1-specific opioid effects. GLP-1 agonists may have distinct analgesic or anti-inflammatory properties that could confound the OtherDM estimate.	The OtherDM signal is a composite. Larger studies should isolate GLP-1, SGLT-2, and other drug classes separately.
Single-database study	Results from one EHR database may not generalise to other healthcare systems or geographic regions.	External validity should be assessed before clinical translation.
Baseline opioid use ascertainment	Baseline opioid use was defined as any dispensation in the 365-day pre-surgery window. This does not distinguish chronic high-dose users from occasional users.	Dose and chronicity of pre-surgical opioid use are important refinements for future analyses.

12. Journal Submission: BMC Musculoskeletal Disorders

BMC Musculoskeletal Disorders (BioMed Central / Springer Nature) is a broad-scope, open-access journal publishing original research on musculoskeletal conditions including orthopaedic surgery outcomes, post-operative complications, and pain management. As a fully open-access journal, all published articles are freely available without subscription, maximising reach for clinical and research audiences.

Category	Details	Action Required
Journal	BMC Musculoskeletal Disorders Publisher: BioMed Central (Springer Nature) ISSN: 1471-2474 (online, open access only)	Confirm current author guidelines at bmcmusculoskeletdisord.biomedcentral.com
Impact Factor	Approximately 2.1-2.5 (2024 JCR) CiteScore: ~3.5 Q2 in Orthopedics	Appropriate for outcomes research. Lower IF than Journal of Arthroplasty but broader scope and open access increases citation potential.
Article Type	Research Article (primary format) No word limit for research articles	Select 'Research Article' at submission. No need to constrain word count.
Word Limit	No word limit for the manuscript body Structured Abstract: maximum 350 words Sections: Background, Methods, Results, Conclusions	Write a comprehensive methods and results section. Take advantage of no word limit to fully describe sensitivity analyses.
Abstract Format	Structured: Background, Methods, Results, Conclusions Maximum 350 words No citations permitted in abstract	All four sections required. Highlight OtherDM aOR and DietControlled protective finding in Results.
Tables & Figures	No limit on tables or figures Figures: TIFF/EPS/PDF, minimum 300 DPI Tables in editable format	Include all sensitivity tables and forest plots. No restriction on supplementary materials.
References	No stated reference limit Vancouver (numbered) format	Comprehensive citation of opioid post-arthroplasty literature, DM pharmacotherapy outcomes, and Firth regression methodology.
Required Statements	Ethics/IRB approval statement (mandatory) Informed consent or waiver statement Data availability statement (mandatory) Conflict of interest declarations Funding statement Authors' contributions section	Complete all mandatory statements before submission. Data availability statement must specify whether data can be shared or explain restrictions.
STROBE Checklist	Required for observational studies Must be submitted as a supplementary file Checklist items should be explicitly addressed in the manuscript	Complete STROBE checklist and upload as Additional File 1.
Article Processing Charge (APC)	USD 2,290 (approximate, 2025 rate) BioMed Central is fully open access — APC is mandatory; no subscription option Waiver/discount available for authors from low-income countries	Confirm current APC on journal website. Check institutional agreements with Springer Nature (many universities have read-and-publish deals covering APC).
Open Access	All articles published under Creative Commons Attribution (CC BY 4.0) Immediately freely available worldwide upon publication Complies with NIH, Wellcome Trust, and most funder open-access mandates	Authors retain copyright. Confirm CC BY 4.0 is acceptable for your institution and funders.
Peer Review	Single-blind or open peer review available (author's choice) Typical first decision: 4-6 weeks Revision turnaround: 4-8 weeks Total time to acceptance: 2-5 months	Consider selecting open peer review if transparency is valued. Prepare a structured response-to-reviewers document.

Category	Details	Action Required
Key Submission Tips	1. Emphasise the pain management and opioid dependency implications — central to BMC MSK scope 2. Highlight the opioid-naïve subgroup (8F) as the most clinically impactful finding 3. Address GLP-1 agonist merging limitation explicitly — high editorial interest 4. Consider adding absolute risk differences (ARDs) alongside ORs for clinical interpretability 5. Include a data availability statement even if data cannot be shared — describe what is available	Draft a Data Availability section. Consider making summary statistics available as a supplementary table to satisfy the statement.
Competing Journals	Alternatives if rejected: - Journal of Orthopaedic Surgery and Research (JOSR, open access, APC ~USD 2,190) - Knee Surgery, Sports Traumatology, Arthroscopy (KSSTA) - The Journal of Arthroplasty (if scope aligns with readmission companion paper) - Pain Medicine (if reframing for pain management audience)	Prepare a tiered submission strategy. JOSR is a logical first alternative.

Method: Firth penalised logistic regression, bootstrap 95% CI (1,000 iterations). Cross-checked with GLM HC3 robust SE. Covariates: age, BMI, HbA1c, diabetes duration, surgery type, HTN, HLD, CAD, depression, baseline opioid use, baseline inpatient visits. GLP-1 receptor agonists merged into OtherDM. Reference: Metformin.